

Some Ideas about Engineering Dielectrics Researches

Shengtao Li¹, Qingquan Lei^{1,2,3,4}

1. State Key Laboratory of Electrical Insulation and Power Equipment, Xi'an Jiaotong University, Shaanxi, China;
2. College of Materials Science and Engineering, Qingdao University of Science and Technology, Qingdao, China;
3. State Key Laboratory of Control and Simulation of Power System and Generation Equipments, Beijing, China;
4. School of Electrical and Electronic Engineering, Harbin University of Science and Technology, Harbin, China;

Abstract- Nanometric dielectrics have always been a focus of engineering dielectrics. Researchers have deepened the understanding of nanometric dielectrics through multi-angle studies. In this paper, we put forward some ideas about engineering dielectrics researches. It is proposed that the name nanodielectric composites should be more precise according to low dimensional physics. For space charge limited current (SCLC), it is considered that SCLC is not solely determined by carriers which are the same as injected carriers in the transition from ohmic region to high field region. In view of pulse related space charge measurement methods, especially pulsed electroacoustic (PEA) method has been mainly adopted, its advantages and disadvantages are described. At last, experimental results of trap-related breakdown and flashover phenomenon, functionalized groups related interfacial compatibility, and classified effects of nanofillers on short-term breakdown and long-term failure are presented to inspire further discussion.

Key words: engineering dielectrics; conductivity; PEA; deep traps; interfacial compatibility; scattering effect

I. INTRODUCTION

Engineering dielectrics [1], especially nanometric dielectrics, have important applications in electrical equipment and electronic devices. The study of the correlation between microscopic, mesoscopic theories and macroscopic performance (3M) is of great significance for the application of engineering dielectrics. Lewis [2] firstly proposed the concept of nanometric dielectric and discovered the excellent insulation properties of it. Due to the large specific surface area of nanoparticles, large quantities of interface regions between the nanofillers and the polymers are formed [3]. It had been widely adopted that the improved performance of nanometric dielectric compared with pure polymers originates from the interface regions. Since the interfacial barrier exhibits the charge carriers' migration in the sample bulk or along the sample surface, the charge transport characteristics can be affected, thus changing the space charge distribution, finally demonstrated as the changed conductivity, electric field distribution, short-term bulk breakdown and surface flashover performance, along with the long-term-corona resistance performance, etc. With continuous development in the field of engineering dielectrics, especially the development of manufacture and characterization methods of nanometric dielectrics, combining with numerical simulation techniques, microscopic and mesoscopic theories have been developed and the issues existed for a long time can be better understood. Some scientific understandings in the field of dielectrics need

to be further strengthened. Around with these issues, we present some issues and show some research progress to inspire further discussion.

II. INTERFACE MODEL

Two interfacial models of nanodielectric composites are commonly adopted, one is the interface of continuous transition region proposed by Lewis [3], and the other is the colloid-like multi-core model proposed by Tanaka [4]. The interface between the nanoparticles and the polymer matrix is in the scale of nm. The interface region of nanometric dielectrics can be approximately described by a continuous two-phase model whose density function and potential function is continuous. The Lewis interface is an ideal interface. The two phases of the interface region do not have significant differences in composition and structure. The Tanaka interface is a solid/liquid interface model similar to a charged colloid, in which the liquid is a continuous electrolyte with isotropic properties, so there is a charge layer in the interface region where a certain charge distribution predominates.

The Lewis interface is an electric double layer formed by the spontaneous transfer of charges caused by band structure difference, especially for the Fermi level, or carrier concentration gradient. The Tanaka interface is an electric double layer formed by spontaneous charge interaction or confinement, which is greatly affected by temperature and ion concentration. For example, when the thermal kinetic energy is greater than the electrostatic potential energy, the electric double layer will disappear, that is, the Tanaka interface is limited by temperature and electrolyte concentration. In contrast, the interfacial structure between inorganic particles and polymer is much complicated, and the electrical, thermal, and mechanical properties between the nanoparticles and polymer matrix are much different, which may cause the complexity and mutability of the interface structure. Therefore, the interface structure between the inorganic particles and polymer is not completely determined by one factor, while would be affected by surface treatment, specific surface area (nanoparticle size), polarity of the polymers, etc. The two models are helpful to understand the improved performance for nanodielectric composites. However, some experimental phenomena cannot be explained well. Such as, the increased crystallinities of surface-treated nanocomposite compared with microcomposites and untreated nanocomposites.

The surface functionalized groups on nanoparticles affect the interaction between the nanofiller and polymer matrix.

Besides, from the perspective of cohesive energy density (CED), the CED of polymer can characterize the intermolecular force between polymer molecules, but also characterize the flexibility of molecular chains. The larger the CED, the better the flexibility of molecular chain is. Hence, it can speculate that the CED of polymeric matrix affects the interfacial bonding strength between polymer matrix and nanoparticles. Moreover, the increased crystallinity could originate from the arranged orderly molecular chains (crystallization zone) in the interface regions.

III. UNDERSTANDING OF CONDUCTIVITY

The forbidden band width of polymeric dielectrics are generally 6~9 eV, which is much bigger than that of inorganic semiconductor with 1~3 eV [5]. It can be considered that the low field conductivity in Ohmic region (below the electric field for Schottky injection, generally about 10 kV/mm) was mainly induced by ion conductance produced by the impurities dissociation of small molecular in polymers. In particular, there is a fundamental difference between electron and ionic conductance in polymers. Electronic conductivity is analyzed according to the energy band model, including free electrons in the conduction band and localized electrons in the forbidden band, in which the former obeys Bloch type wave function, and the latter obeys Anderson type wave function. The electrons in the insulators are subject to the Fermi-Dirac quantum distribution, and the ions are subject to the Maxwell-Boltzmann classical distribution. There are no free ions in the dielectrics, while ions can overcome a potential barrier and generate hopping conduction under the action of an electric field. According to the above characteristics, generally, electronic conductivity dominates at high electric field and low temperature, whilst the ionic conductivity dominates at low electric field and high temperature. Therefore, both electrons and ions participate in the conductivity process in polymers, in which the electrons mainly originate from the charge injection from metal electrode, and the ions originate from impurities dissociation in polymers.

IV. PULSED ELECTROACOUSTIC (PEA) METHOD

The most widely adopted method for measuring the space charge of solid dielectrics is the pulsed electroacoustic method (PEA) proposed by T. Takada [6-7]. PEA has been applied to the measurement of space charge for solid dielectrics and cables. The space-time distribution of space charge can be displayed with a resolution of μm . It can be used to study the relationship between the aging of insulating materials and the accumulation and distribution of space charge. According to the test principle of PEA, it reflects the charge and electric field distribution characteristics inside the sample when the electric field is applied, whilst cannot clearly distinguish the type of charges (electron or ion). Besides, based on merely the signal measurement of piezoelectric sensor, the space charge distribution and the corresponding electric field distribution are obtained by convolution method, which lacks the understanding of the nature of the signal.

V. SOME PROGRESSES IN ENGINEERING DIELECTRICS

Generally, trap characteristics of pure polymers can be modulated by incorporating nanofillers with different filler contents [8-9] and different surface functionalization groups [10], irradiation [11], and blending [12], etc. In our previous work, the correlation between traps characteristics and insulation failure was analyzed [13-14]. Besides, results of thermally stimulated depolarization current (TSDC) and surface potential decay (SPD) show that incorporating with a low filler contents into polymers can increase the trap depth, which could be derived from the independent interaction between nanoparticles and polymers [13]. The enhanced trapping effect of deep trap can effectively reduce mean free path of carriers and the energy obtained from electric field [15]. Consequently, the short-term insulation failure, including the breakdown and surface flashover, increases as an increase of deep trap energy due to the effective suppression of carrier mobility and secondary electrons emission (SEE) [11,16] and its multiplication process.

Incorporating polymers with nanofillers is a preferential method to improve electrical performance, while functionalization on filler surface has always been employed. A more fundamental and refined understanding of the interfacial effect on electrical properties indicated that the compatibility in the interface is closely related to breakdown performance [10]. The improved breakdown strength corresponds with the composite possessing better intermolecular compatibility and deeper traps. According to molecular simulation, it is further confirmed that surface functionalized groups can modulate interfacial compatibility and trap distribution by intermolecular interaction energy and electrostatic potential [10], thus affecting charge carriers migration in composites and finally resulting in the changed electrical properties of it.

In addition to the changed trapping effect in nanodielectric composites induced by interface region, the classified effects of nanofillers on short-term breakdown and long-term corona-resistance for nanocomposites was proposed [9]. The interesting phenomenon of different eroded surface morphologies and the layered rings occurred on the surface of nanocomposites [9] for high loading composites inspire further exploration. Owing to the violent discharge on sample surface, the electrostatic Coulomb force or interfacial potential barrier around the nanoparticles is considered ineffective in shielding from the bombardment by the continuously energetic charged ions [9]. Compared with the improved breakdown performance at low filler contents, the corona-resistance of nanocomposites with higher filler content exhibits longer failure time, a corona-resistance model for nanocomposites is proposed on the basis of synergistic effects of energy dissipation by nanoparticle scattering and charge dissipation. Scattering effect of nanoparticle arises from the 'hard' collision with the energetic ions. The area below the needle tip where suffers the most erosion is therefore protected. The scattering effect against charged ions plays an important role in energy dissipation, and faster charge dissipation induced by shallower traps mitigates energy concentration. Since the small charge

mobility and charge energy, traps energy with a few eV can play effective role in trapping charges during short-term breakdown, whilst it is ineffective in the violent corona-resistance process.

VI. CONCLUSION

This paper expounds the concept of nanometric dielectrics around with its interfacial structures, and points out the issues that need to be considered, including space charge limited current (SCLC) and the space charge measurement by PEA method. Accurate understanding of nanometric dielectrics and the interface between inorganic fillers and polymers, and the differences between electronic conductivity and ionic conductivity are of great significance for the research of improving performance of nanometric dielectrics. PEA method is an effective means for space charge measurement of solid dielectrics, but should pay attention to its disadvantages. Trapping effect, interfacial compatibility, and classified effects of nanofillers on short-term breakdown and flashover, and long-term failure are presented to enlighten further thinking. The road ahead is long and has no ending; yet high and low, we will search with our will unbending.

ACKNOWLEDGMENT

This work is supported by National Basic Research Program of China (973 Program) under Project with No. 2015CB251003, and the National Natural Science Foundation of China (NSFC) under Project with No. 51337008.

REFERENCES

- [1] Lei Q, Li S. Several important issues and thinking about engineering dielectrics [J]. *High Voltage Engineering*, 2015, 41(8): 2473-2480.
- [2] Lewis T J. Nanometric dielectrics[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 1994, 1(5): 812-825.
- [3] Lewis T J. Interfaces are the dominant feature of dielectrics at the nanometric level[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 2004, 11(5): 739-753.
- [4] Tanaka T, Kozako M, Fuse N, et al. Proposal of a multi-core model for polymer nanocomposite dielectrics[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 2005, 12(4): 669-681.
- [5] Teyssedre G. Charge Transport Modeling in Insulating Polymers : From Molecular to Macroscopic Scale[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 2005, 12(5):857-875.
- [6] Takada T, Sakai T. Measurement of electric fields at a dielectric/electrode interface using an acoustic transducer technique[J]. *IEEE Transactions on Electrical Insulation*, 1983 (6): 619-628.
- [7] Li Y, Yasuda M, Takada T. Pulsed electroacoustic method for measurement of charge accumulation in solid dielectrics[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 1994, 1(2):188-195.
- [8] Li S, Yin G, Bai S, et al. A new potential barrier model in epoxy resin nanodielectrics[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 2011, 18(5): 1535-1543.
- [9] Xie D, Li S, Min D, et al. Classified Effects of Nanofillers on DC Breakdown and Partial Discharge Resistance of Polypropylene/Alumina Nanocomposites[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*. (Accept)
- [10] Li S, Xie D, Qu G, et al. Tailoring Interfacial Compatibility and Electrical Breakdown Properties in Polypropylene based Composites by Surface Functionalized POSS[J]. *Applied Surface Science*, 2019, 478(6): 451-458.
- [11] Huang Y, Min D, Li S, et al. Surface flashover performance of epoxy resin microcomposites improved by electron beam irradiation[J]. *Applied Surface Science*, 2017, 406: 39-45.
- [12] Zhou Y, He J, Hu J, et al. Evaluation of polypropylene/polyolefin elastomer blends for potential recyclable HVDC cable insulation applications[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 2015, 22(2): 673-681.
- [13] Li S. Charge dynamics: Linking traps to insulation failure [C]//2015 IEEE 11th International Conference on the Properties and Applications of Dielectric Materials (ICPADM), 2015: 1-14.
- [14] Li S, Min D, Wang W, et al. Linking Traps to Dielectric Breakdown through Charge Dynamics for Polymer Nanocomposites[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 2016, 23(5):2777 – 2785.
- [15] Wang W, Min D, Li S. Understanding the conduction and breakdown properties of polyethylene nanodielectrics: effect of deep traps [J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 2016, 23(1): 564-572.
- [16] Li S, Nie Y, Wang W, et al. Surface flashover performance of phenolphthalein modified LDPE in vacuum[J]. *IEEE Transactions on Dielectrics and Electrical Insulation*, 2016, 23(5): 3215-3223.